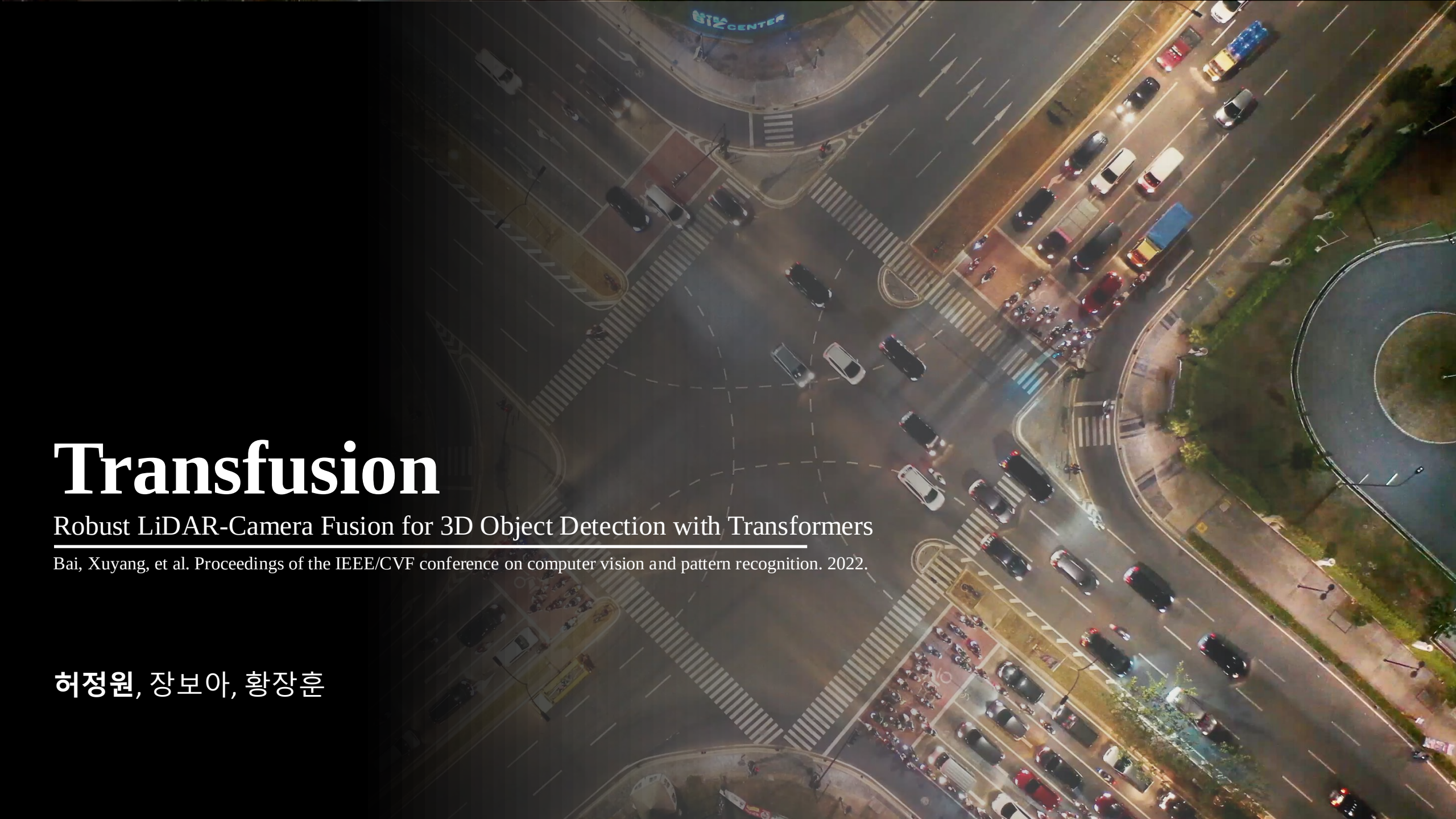




Transfusion

Robust LiDAR-Camera Fusion for 3D Object Detection with Transformers

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1. Introduction

What is 3D Object Detection?

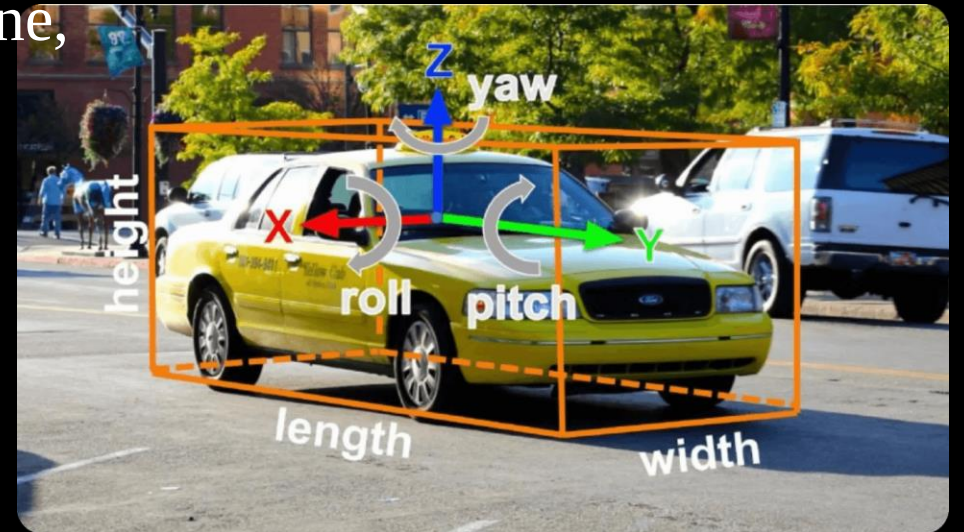
3D object detection aims to localize a set of objects in 3D space and recognize their categories.

$$\mathcal{B} = f_{det}(\mathcal{I}_{sensor}),$$

$\mathcal{B} = \{B_1, \dots, B_N\}$ is a set of N 3D object in a scene,

f_{det} is a 3D object detection model,

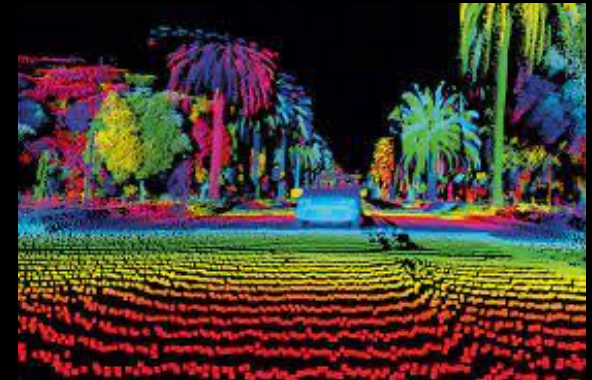
$\mathcal{I}_{sensor}$ is one or more sensory inputs.



Sensory Inputs

Radars, cameras, and LiDAR (Light Detection And Ranging) sensors are the three most widely adopted sensory types

- Radar: Long detection range and robust to weather conditions. Velocity(Doppler)
- Camera: Cheap and easily accessible and crucial for understanding semantics.
- LiDAR: Accurate 3D information directly acquired by LiDAR sensors.



Radar, Camera, LiDAR

Multi-modal detector

- LiDAR provide accurate depth information.
- But, LiDAR-only methods are surely insufficient for robust 3D detection due to the sparsity of point clouds.



VoxFormer: <https://github.com/NVlabs/VoxFormer>

Small or distant objects are difficult to detect in LiDAR modality.

In contrast, such objects are still clearly visible and distinguishable in high-resolution images.

→ **Utilizing the best of the two worlds, i.e., multi-modal detectors.**

Problems

- Point-level fusion methods suffer from two major problems.
Element-wise addition or concatenation
 - The performance **degrades seriously with low-quality image features**
 - Finding the hard association between sparse LiDAR points and dense image pixels.
 - **wastes many image features**
 - **heavily relies on high-quality calibration** between two sensors.



Figure 1. Left: An example of bad illumination conditions. Right: The sparsity of point clouds

Transfusion

- Reposition the focus of the fusion process.

From hard-association to soft-association, leading to **the robustness against degenerated image quality and sensor misalignment.**

1. Reveal a crucial aspect to robust fusion, namely, the **soft-association** mechanism.
2. Propose a novel **transformer-based** LiDAR-camera fusion model for 3D detection.
3. An **image-guided query** initialization module boost the quality of initial bounding box predictions for image fusion and handle objects that are hard to detect in point clouds.
4. The state-of-the-art 3D detection performance on nuScenes.

2. Methodology

Overall pipeline of Transfusion

1. A **LiDAR BEV feature map** and an **image feature map** from convolutional backbones.
2. Transformer-based detection head.
 - a) First **decodes object queries** into initial bounding box predictions using the LiDAR information.
 - b) Performs **LiDAR-camera fusion** by attentively fusing object queries with useful image features.

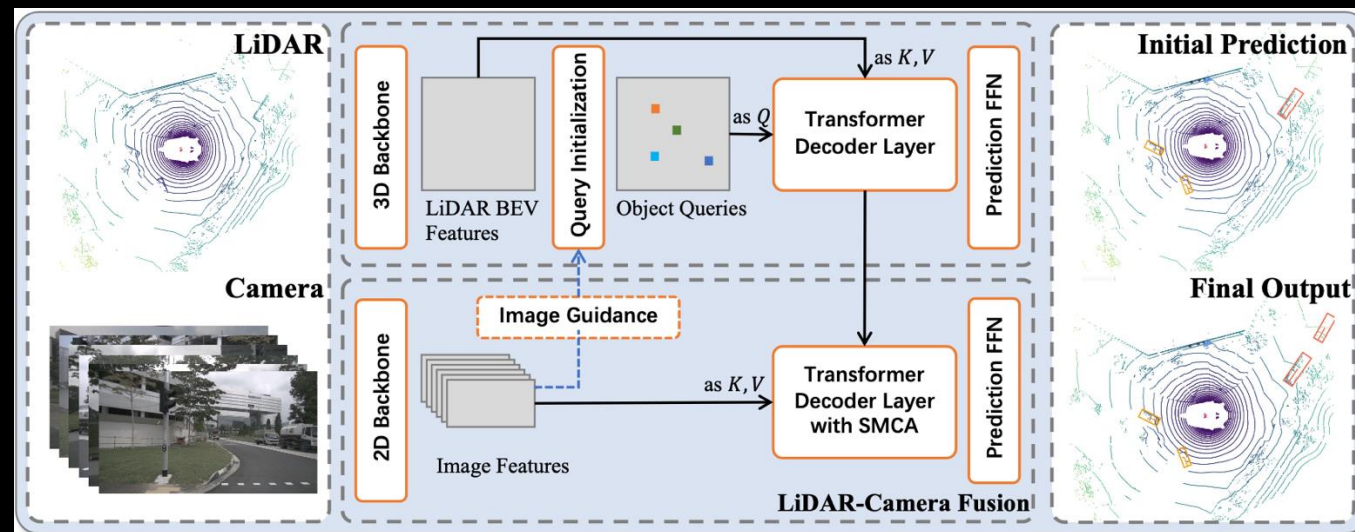


Figure 2. Overall pipeline of TransFusion.

Query Initialization

- Predict a **class-specific heatmap** $X \times Y \times K$

$X \times Y$ describes the size of the BEV feature map

K is the number of categories

- Regard the heatmap as $X \times Y \times K$ object candidates and **select the top-N candidates** for all the categories.

→ **Initial object queries** will locate at or close to the potential object centers, eliminating the need of multiple decoder layers to refine the locations.

Make the object queries **category-aware** by equipping each query with a **category embedding**.

It serves as a useful side information when modelling the **object-object relations** in the self-attention modules and the **object-context relations** in the cross-attention modules.

Transformer Decoder and FFN

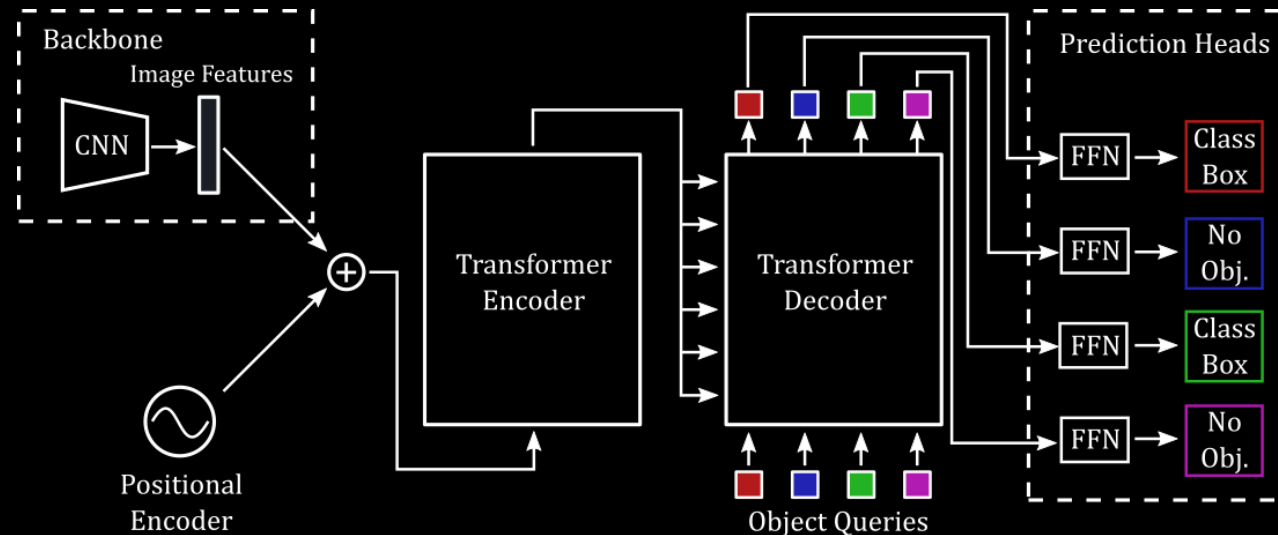
- The decoder layer follows the design of DETR

The **cross attention between object queries and the feature maps**.

The **query positions are embedded** and element-wisely summed with the query features.

→ Enables the network to reason about both **context and position** jointly.

The N object queries containing rich instance information are then **independently decoded into boxes** and class labels by a feed-forward network (FFN). Following Center-Point.



3. Experiments

Main results

Method	Modality	Voxel Size (m)	mAP	NDS	Car	Truck	C.V.	Bus	Trailer	Barrier	Motor.	Bike	Ped.	T.C.
PointPillar [14]	L	(0.2, 0.2, 8)	40.1	55.0	76.0	31.0	11.3	32.1	36.6	56.4	34.2	14.0	64.0	45.6
CBGS [68]	L	(0.1, 0.1, 0.2)	52.8	63.3	81.1	48.5	10.5	54.9	42.9	65.7	51.5	22.3	80.1	70.9
CenterPoint [57]	L	(0.075, 0.075, 0.2)	60.3	67.3	85.2	53.5	20.0	63.6	56.0	71.1	59.5	30.7	84.6	78.4
PointPainting [46]	LC	(0.2, 0.2, 8)	46.4	58.1	77.9	35.8	15.8	36.2	37.3	60.2	41.5	24.1	73.3	62.4
3D-CVF [59]	LC	(0.05, 0.05, 0.2)	52.7	62.3	83.0	45.0	15.9	48.8	49.6	65.9	51.2	30.4	74.2	62.9
PointAugmenting [47]	LC	(0.075, 0.075, 0.2)	66.8	71.0	87.5	57.3	28.0	65.2	60.7	72.6	74.3	50.9	87.9	83.6
MVP [58]	LC	(0.075, 0.075, 0.2)	66.4	70.5	86.8	58.5	26.1	67.4	57.3	74.8	70.0	49.3	89.1	85.0
FusionPainting [51]	LC	(0.075, 0.075, 0.2)	68.1	71.6	87.1	60.8	30.0	68.5	61.7	71.8	74.7	53.5	88.3	85.0
TransFusion-L	L	(0.075, 0.075, 0.2)	65.5	70.2	86.2	56.7	28.2	66.3	58.8	78.2	68.3	44.2	86.1	82.0
TransFusion	LC	(0.075, 0.075, 0.2)	68.9	71.7	87.1	60.0	33.1	68.3	60.8	78.1	73.6	52.9	88.4	86.7

Table 1. Comparison with SOTA methods on the nuScenes test set.

Ablation Studies

- TransFusion-L does not require more decoder layers.
- The image feature fusion and image-guided query initialization bring 4.8% and 1.6% mAP gain.

	C.A.	I.D.	#Layers	#Epochs	mAP	NDS
a)	✓	✓	1	12	60.0	66.8
b)		✓	1	12	54.3	63.9
c)	✓	✓	3	12	59.9	67.1
d)	✓		1	12	24.0	33.8
e)	✓		3	12	28.3	43.4
f)	✓		3	36	46.9	57.8

Table 6. Ablation of the query initialization module.
C.A.: category-aware; I.D.: input-dependent.

	mAP	NDS	Params (M)	Latency (ms)
CenterPoint	57.4	65.2	8.54	117.2
TransFusion-L	60.0	66.8	7.96	114.9
CC	63.3	67.6	8.01 + 18.34	212.3
PA	64.2	68.7	13.9 + 18.34	288.2
w/o Fusion	61.6	67.4	9.08 + 18.34	215.0
w/o Guide	64.8	69.3	8.35 + 18.34	236.9
TransFusion	65.6	69.7	9.47 + 18.34	265.9

Table 7. Ablation of the proposed fusion components.

4. Conclusion

Conclusion

- An **effective and robust transformer-based** LiDAR-camera 3D detection framework with a **soft-association** mechanism to adaptively determine where and what information should be taken from images.

Questions?